Sujay Patil

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Education

University of Glasgow **September 2021 – July 2026**

Integrated Master of Science (MSci) in Computer Science *Glasgow, Scotland, UK*

Relevant Courses: Algorithmics, Networks and Operating Systems, Robotics, Database Systems, Machine Learning, Deep Learning, Systems Programming, Cyber Security, Big Data, Web Application Development, Start-up Growth Engineering

Experience

Glasgow University Software Service **Mar 2024 – Mar 2026**

Software Engineer *Glasgow, Scotland, UK*

- Translated Analogue Circuit Design Tuning MatLab Codes into Production Python Code using **ML algorithms.**
- Implemented **Machine Learning Techniques** to build and train **neural networks** for **constraint optimization.**
- Deployed **licensed proprietary software** compatible across multiple **Linux** distributions via a custom GUI installer.
- Built test cases to guarantee the optimisation algorithm's accuracy and efficiency.
- Technologies used: **NumPy**, **PyTorch**, **MATLAB**

Barclays **Jun 2025 – Aug 2025**

Technology Analyst Intern *Glasgow, Scotland, UK*

- Operated as a **Risk & Control Partner** within the **Markets Pre-Trade Technology Team**.
- Created a comprehensive list of annual **supplier contractual obligation waivers** across 5 different risk horizontals.
- Reviewed the **sub-limited risk JIRA portfolio** to assess completion status and risk treatment for outstanding items.
- Identified **Cyber waiver tickets** to raise them to limited risk status in **ORAC** to enable tracking and oversight.
- Automated a list, that **maps** all Standard and Horizontal Owners to their respective areas of responsibility.
- Technologies used: **ORAC**, **Jira**, **Microsoft Excel**, **Microsoft Teams**

National Manufacturing Institute Scotland **Jun 2024 – Jun 2025**

Software Engineer Intern *Paisley, Scotland, UK*

- Developed an **AI system** that can read, extract, and classify invoice data to carry out complete carbon accounting.
- Integrated all **Scope 3 emission categories** based on the GHG protocol, alongside full carbon accounting functions.
- Built a **Machine Data Analysis** system for **80+ machines** with real-time energy insights and automated reporting.
- Developed an **interactive dashboards** to highlight load imbalance, idle consumption, and batch-level cost metrics.
- Technologies used: **Python**, **Django**, **React.js**, **SQL**, **OpeanAI API**, **Figma**, **Git**, **Microsoft Excel**

GU Orbit **Sep 2022 – Jun 2025**

Head Software Engineer *Glasgow, Scotland, UK*

- Developed the prototype of the **processing pipeline** to be deployed on an **aerial testing platform**.
- Developed Python and C++ scripts and interfaced **BMP180 sensors** to **Rasberry Pi** with **i2c connection**.
- Successfully completed the **CloudView balloon launch**. The trip lasted 3 hours and sent back large amounts of data.
- Technologies used: **C++**, **Python**, **Docker**, **Raspberry Pi**, **Git**, **Microsoft Teams**

Projects

AR-Assisted Sign Language Learning | *PyTorch, React.js, MediaPipe, ONNX Runtime Web* **Sep 2025 - Apr 2026**

- For my Master's project, I researched **AR-assisted sign language education** by developing a browser based **BSL learning platform** with real-time gesture recognition and feedback. The system combined MediaPipe hand tracking, neural networks, and DTW based motion analysis to evaluate static and dynamic gestures, and was evaluated through a user study on engagement, usability, and accessibility.

AI-Powered Life Cycle Assessment Tool - LCAi | *Django REST, React,js, OpenAI API* **Sep 2024 - Mar 2025**

- In collaboration with the National Manufacturing Institute Scotland, I developed an AI-powered Lifecycle Assessment (LCA) tool designed to evaluate the **environmental impact** of products. The tool automatically analyzes **Bills of Materials (BOMs)** and related product data to assess environmental impacts across the product's entire lifecycle.

UofG Gaming Lab - Project Gundam | *Unity, C#, Google Cloud* **Sep 2023 - Mar 2024**

- We developed a **Mobile AR application** for the **WorldCon 2024** event, using Unity. To encourage attendees, to explore Glasgow, this app aims to enhance visitors' experiences of the city. In addition the app also provides the user with a compass and a Gundam tracker, The app spawns several sorts of Gundam Models across Glasgow.

Technical Profile

Languages: Python, Java, C, C++, C#, HTML, CSS, JavaScript, TypeScript, SQL, MATLAB

Frameworks: Django REST, Flask, React.js, jQuery, NumPy, Unity, PyTorch, ROS (Robot Operating System)

Technical skills: Git, Full Stack Development, Artificial Intelligence, Mobile AR Development, Object Oriented Programming, Human Computer Interaction, Data Engineering, Project Management, Documentation, LaTex, Risk Analysis, Data Analysis, Cloud Computing, Deep Learning, Business Development